Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science)

Semester – V

CDS-354
Lab on CDS-351
Digital Forensics-01 Laboratory

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Analyze a given disk image using Autopsy and list the recovered files, including any hidden or deleted files. **[15M]**

Q2 Perform a basic forensic investigation on a Windows machine using FTK Imager to acquire memory and explain the steps. **[10M]**

Q3 Viva **[5M]**

Q4 Lab Book **[5M]**

Record the steps for disk image analysis using a forensic tool.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use Wireshark to capture network traffic and identify any suspicious activity, such as potential data exfiltration. **[15M]**

Q2 Examine an email header to trace its origin and identify any signs of spoofing. **[10M]**

Q3 Viva **[5M]**

Questions on packet analysis and identifying common network threats.

Q4 Lab Book. **[5M]**

Record the steps for performing a packet capture using Wireshark.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Recover deleted files from a USB drive using Test Disk and explain the process. **[15M]**

Q2 Perform an analysis of a web browser's history to identify user activity using the browser's forensic tool. **[10M]**

Q3 Viva **[5M]**

Questions on file recovery methods and the importance of hash values in forensics.

Q4 Lab Book **[5M]**

Document the process of file recovery from a storage device.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use bulk_extractor to extract information such as credit card numbers or email addresses from a disk image. **[15M]**

Q2 Perform metadata analysis on a PDF file to determine its creation details and any potential tampering. **[10M]**

Q3 Viva **[5M]**

Questions about metadata analysis and its importance in forensics.

Q4 Lab Book **[5M]**

Record the process of extracting sensitive information using bulk_extractor.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Conduct a forensic analysis on a mobile device image using MOBILedit, identifying contacts, messages, and media files. **[15M]**

Q2 Analyze a downloaded malware sample and identify the network connections it attempts to establish using a sandbox environment. **[10M]**

Q3 Viva **[5M]**

Questions about mobile device forensics and network forensics.

Q4 Lab Book **[5M]**

Record the analysis process for mobile device data recovery.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Analyze a compromised system for signs of unauthorized access using event logs and identify the attacker's activities. **[15M]**

Q2 Perform a hash comparison of two files using a hashing tool (e.g., MD5 or SHA256) and explain why hashing is important in forensics. **[10M]**

Q3 Viva **[5M]**

Questions about log analysis and hash functions.

Q4 Lab Book **[5M]**

Record the process of investigating a compromised system using event logs

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use Sleuth Kit (TSK) to analyze file system structures and retrieve deleted files. **[15M]**

Q2 Create a forensic image of a hard disk using dd and verify its integrity. **[10M]**

Q3 Viva **[5M]**

Questions on file system forensics and disk imaging.

Q4 Lab Book **[5M]**

Document the steps of using TSK for file system analysis.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use Volatility to perform memory analysis on a captured RAM image and identify any running processes or malware. **[15M]**

Q2 Examine web server logs for signs of a web attack (e.g., SQL injection or brute force). **[10M]**

Q3 Viva **[5M]**

Questions on memory forensics and web server log analysis.

Q4 Lab Book **[5M]**

Document memory analysis using Volatility

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Recover and analyze internet history, cookies, and cached files from a web browser using specialized browser forensic tools. **[15M]**

Q2 Perform steganalysis on an image to identify hidden messages or files using a steganography detection tool. **[10M]**

Q3 Viva **[5M]**

Questions about steganography and browser forensics.

Q4 Lab Book **[5M]**

Record the process of internet history recovery.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Perform keyword search analysis on a disk image using Autopsy to identify files related to a specific investigation. **[15M]**

Q2 Identify the system's boot process using log files and analyze any tampering or modifications. **[10M]**

Q3 Viva **[5M]**

Questions about keyword search techniques in forensic investigations.

Q4 Lab Book **[5M]**

Document the steps of keyword search in forensic analysis.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Extract and analyze EXIF data from images to determine location and camera information. **[15M]**

Q2 Perform timeline analysis on a file system to determine when files were created, modified, and deleted. **[10M]**

Q3 Viva **[5M]**

Questions about EXIF data and timeline analysis.

Q4 Lab Book **[5M]**

Document EXIF data extraction.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use a forensic toolkit to analyze the contents of a virtual machine disk image. **[15M]**

Q2 Perform OS fingerprinting and analyze the results to identify system vulnerabilities. **[10M]**

Q3 Viva **[5M]**

Questions on virtual machine forensics and OS fingerprinting.

Q4 Lab Book **[5M]**

Record the analysis of a virtual machine disk image.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use Wireshark to analyze network traffic for signs of a Denial of Service (DoS) attack. **[15M]**

Q2 Identify and recover deleted browser bookmarks using browser forensic tools. **[10M]**

Q3 Viva **[5M]**

Questions about detecting network-based attacks and browser forensics.

Q4 Lab Book **[5M]**

Document the detection of network anomalies using Wireshark.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Perform registry analysis using RegRipper to identify recently accessed files on a Windows system. **[15M]**

Q2 Examine log files from a Linux system to identify unauthorized login attempts. **[10M]**

Q3 Viva **[5M]**

Questions about registry analysis and Linux log analysis.

Q4 Lab Book **[5M]**

Document registry analysis for user activity tracking.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use FTK Imager to mount a disk image and explore its contents for evidence. **[15M]**

Q2 Analyze email archives for any signs of phishing or malicious attachments. **[10M]**

Q3 Viva **[5M]**

Questions about disk image analysis and email forensics.

Q4 Lab Book **[5M]**

Document the process of mounting and analyzing a disk image.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Analyze Windows event logs to identify system shutdowns, logins, and potential

misuse. **[15M]**

Q2 Use OSForensics to compare file hashes and identify any changes or tampered files. **[10M]**

Q3 Viva **[5M]**

Questions about Windows event logs and file integrity checking.

Q4 Lab Book **[5M]**

Document event log analysis for system activity tracking.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Extract data from a cloud-based service like Google Drive using forensic methods. **[15M]**

Q2 Use FTK to search for keywords in a disk image and present the findings. **[10M]**

Q3 Viva **[5M]**

Questions about cloud forensics and keyword search in forensic investigations.

Q4 Lab Book **[5M]**

Record the steps for cloud data extraction.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use Autopsy to generate a report from a disk image, detailing findings like file metadata and deleted files. **[15M]**

Q2 Examine a chat log to uncover communications related to a specific incident. **[10M]**

Q3 Viva **[5M]**

Questions about report generation in forensic tools.

Q4 Lab Book **[5M]**

Document the generation of forensic reports.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Analyze a memory dump using Volatility to detect malware or other suspicious processes. **[15M]**

Q2 Use Autopsy to recover and analyze USB device usage on a computer. **[10M]**

Q3 Viva **[5M]**

Questions about memory forensics and USB device analysis.

Q4 Lab Book **[5M]**

Document the analysis of a memory dump for malware detection

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use bulk_extractor to extract URLs and email addresses from a forensic disk image. **[15M]**

Q2 Analyze the contents of a SQLite database from a mobile application to retrieve user activity. **[10M]**

Q3 Viva **[5M]**

Questions about SQLite forensics and data extraction using bulk_extractor.

Q4 Lab Book **[5M]**

Record steps for extracting data from databases.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Conduct a steganographic analysis on multiple files to find hidden data. **[15M]**

Q2 Perform timeline analysis using a forensic tool to reconstruct events on a disk. **[10M]**

Q3 Viva **[5M]**

Questions about timeline analysis and steganography detection.

Q4 Lab Book **[5M]**

Record steganographic detection procedures.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use OSForensics to extract and analyze temporary internet files. **[15M]**

Q2 Recover data from a corrupted document file using forensic recovery techniques. **[10M]**

Q3 Viva **[5M]**

Questions about internet file forensics and corrupted file recovery.

Q4 Lab Book **[5M]**

Document the analysis of temporary internet files.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use forensic software to recover deleted partitions from a hard disk. **[15M]**

Q2 Perform analysis on system logs to track changes in user accounts and permissions. **[10M]**

Q3 Viva **[5M]**

Questions about partition recovery and log analysis.

Q4 Lab Book **[5M]**

Record steps for partition recovery using a forensic tool.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Use Recuva or any file recovery tool to recover deleted files from a storage device. **[15M]**

Q2 Perform analysis on a browser cache to uncover visited sites and downloaded files. **[10M]**

Q3 Viva **[5M]**

Questions about file recovery and browser cache analysis.

Q4 Lab Book **[5M]**

Document the steps for file recovery from storage media.

Savitribai Phule Pune University
T.Y. B.Sc (Cyber and Digital Science) Sem-V
(CDS-354 Lab on CDS-351- Digital Forensics-01)

Time : 3 Hours **Max. Marks: 35**

Q1 Extract and analyze data from an encrypted archive file using forensic tools. **[15M]**

Q2 Investigate suspicious Windows processes using Process Explorer and document any anomalies. **[10M]**

Q3 Viva **[5M]**

Questions about encryption and forensic analysis of processes.

Q4 Lab Book **[5M]**

Record steps for decrypting and analyzing an archive.